

# AreaSection: Simultaneous Population and Land-Area Balance in Minimum-Edge-Cut Redistricting (Revision 1)

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## Abstract

Standard minimum-edge-cut redistricting (and its ratio-optimal extension GeoSection, T.1) treats land area as irrelevant to bisection: the algorithm cares only about minimising boundary length subject to population balance. This produces a systematic pattern — *urban peeling* — in which the first bisection level always separates the densest metropolitan core from the rest of the state, because a compact urban boundary is cheapest. Urban peeling is politically contentious: it concentrates Democratic voters into a single compact district at every recursion level, producing implicitly packed maps even with zero partisan input.

We introduce **AreaSection**, a dual-constraint extension that adds a second METIS balance constraint requiring each half to receive approximately 50 % of the state’s land area ( $\pm 10$  % swing allowed). The area constraint is implemented via METIS’s `ncon=2` multi-constraint partitioning with per-constraint imbalance vectors (`ubvec`). Ratio selection uses the same isoperimetric normalisation as GeoSection ( $EC/\sqrt{\min(i, k-i)}$ ) to prevent the caterpillar pathology.

A Lorenz-curve pre-filter identifies ratios where simultaneous population-and-area balance is geometrically infeasible and skips them, preventing wasted METIS calls.

Applying AreaSection to all 50 states with 50 seeds per ratio, 44 states converge cleanly. The top-level natural ratio shifts dramatically versus GeoSection: Illinois moves from 1:16 to 8:9, New York from 1:25 to 13:13, Florida from 1:27 to 14:14. States with genuinely concentrated populations (Georgia: 3:11, Massachusetts: 2:7) retain asymmetric splits because the area constraint cannot be satisfied at more balanced ratios while maintaining contiguous regions.

A head-to-head comparison across 34 states (all states with complete data for both algorithms) shows that AreaSection and GeoSection produce *identical seat counts* in 25 of 34 states (76 %). In the 9 states that differ, 5 favour GeoSection by 1 seat and 4 favour AreaSection by 1 seat (net:  $-1$  Democratic seat under AreaSection). There is no systematic partisan direction. Within the states that match, AreaSection produces a cleaner competitiveness profile: GeoSection’s urban-peeling strategy creates knife-edge suburban districts near 49 % Democratic vote share as a geometric byproduct; AreaSection eliminates these. AreaSection is a geometric-regularity mechanism, not a partisan mechanism: it changes the bisection tree structure without systematically changing partisan seat allocations. All 8 competitive states (46–55 % Democratic

vote) show Republican over-representation under both algorithms (mean  $-6.2$  pp gap), confirming that geographic voter sorting — not the bisection algorithm — drives the proportionality deficit.

# 1 Introduction

Minimum-edge-cut recursive bisection produces congressional district maps by repeatedly splitting the state’s census-tract adjacency graph into two population-balanced halves, minimising the total shared boundary length at each level. The algorithm makes no reference to partisan data: it sees only geography and population. Yet on states with concentrated metropolitan areas, it reliably produces a specific pattern:

*The first bisection peels off the densest urban core as a single compact district, leaving the rest of the state to be bisected at subsequent levels.*

We call this **urban peeling**. It arises not from any partisan intent but from a geometric fact: the boundary surrounding a dense city is shorter than any boundary that bisects the state into two comparably-populated regions. Illinois splits 1:16 at the first level (Chicago peel), not 8:9. New York splits 1:25 (New York City peel), not 13:13. Wisconsin splits 1:7 (Milwaukee peel), not 4:4.

GeoSection (T.1) adds isoperimetric normalisation to prevent the worst form of this pathology — the caterpillar, where 1:k-1 wins at every single level regardless of state shape — but it does not eliminate urban peeling for states where the urban core is genuinely compact. A normalised 1:7 split in Wisconsin still wins when Milwaukee is as compact as it is: the normalisation says “this boundary is as short as a 1:7 boundary could be,” which does not tell us whether it is shorter than a 4:4 boundary.

Urban peeling is legally contentious for two reasons. First, it concentrates a disproportionate share of the minority party’s voters into a single safe district at every recursion level, producing implicitly packed maps. Second, it creates near-toss-up districts in the suburban ring around the peeled city — a predictable byproduct of the boundary geometry — that can influence which party controls the chamber. Both effects arise with zero partisan input, but they are not politically neutral.

**Our contribution.** We introduce **AreaSection**, which adds a second METIS balance constraint requiring each half of every bisection to receive approximately 50% of the state’s land area, with a  $\pm 10\%$  swing tolerance. The area constraint is enforced simultaneously with the population constraint using METIS’s native dual-constraint (`ncon=2`) partitioning with per-constraint imbalance vectors (`ubvec`).

The area constraint cannot be satisfied by a 1:16 split in Illinois because Chicago occupies  $< 5\%$  of the state’s land. A near-equal population split is therefore geometrically forced by the area balance requirement. Illinois moves from 1:16 (GeoSection) to 8:9 (AreaSection). New York moves from 1:25 to 13:13. Florida moves from 1:27 to 14:14.

We make three contributions:

1. **The AreaSection algorithm** (§3): dual population+area METIS bisection with isoperimetric ratio normalisation and a Lorenz-curve feasibility pre-filter.

2. **50-state empirical sweep** (§4): natural ratio results for all 50 states, showing which states are forced to near-equal splits and which retain asymmetric splits due to geographic concentration.
3. **Partisan comparison** (§4): head-to-head GeoSection vs. AreaSection on Wisconsin and North Carolina, showing identical seat allocations but different competitiveness profiles.

**Scope and relationship to T.1.** AreaSection is a companion paper to GeoSection (T.1). Both algorithms use ratio scanning with isoperimetric normalisation. The difference is the vertex weight model: GeoSection uses `ncon=1` (population only); AreaSection uses `ncon=2` (population + land area). All other algorithmic infrastructure — recursive bisection tree, orientation-aware subgraph splitting, proportionality analysis — is shared.

AreaSection does not incorporate partisan data. It is a purely geographic constraint: land area is a federal public record (TIGER ALAND field), computed mechanically, with no interpretive degrees of freedom. *We minimised edge-cut subject to geographic balance and verified the result is not implicitly partisan.* This is a stronger and more legally durable claim than minimising a partisan metric directly.

## 2 Related Work

**Minimum-edge-cut redistricting.** The algorithmic redistricting literature has converged on edge-cut minimisation as a mathematically clean, politically neutral criterion [Chikina et al., 2017, DeFord et al., 2019, Duchin, 2020]. METIS multilevel graph partitioning [Karypis and Kumar, 1998] is the standard solver. Della-Libera [2026a] establish that for states where the seed-sweep converges, the MEC plan is a property of the graph, not the seed. GeoSection [Della-Libera, 2026b] introduces isoperimetric normalisation to select among feasible split ratios at each bisection level.

**Compactness and geographic fairness.** Polsby-Popper [Polsby and Popper, 1991], Reock [Reock, 1961], and convex-hull ratio are standard single-district compactness metrics used in court challenges. These metrics evaluate the shape of individual districts, not the structure of the bisection tree. Edge-cut minimisation is related but distinct: it minimises total boundary length across all districts simultaneously, which is a stronger and more globally consistent criterion.

Area balance as a redistricting criterion has received limited formal attention. Vickrey [1961] proposes that legislative districts should be “compact and contiguous” without specifying land area formally. Niemi et al. [1990] study compactness empirically but focus on shape metrics. To our knowledge, AreaSection is the first algorithm to enforce simultaneous population and land-area balance as a formal METIS dual constraint.

**Multi-constraint graph partitioning.** METIS supports  $k$ -way partitioning with `ncon > 1` balance constraints [Karypis and Kumar, 1999]. The dual-constraint mode has been used in scientific computing for load balancing with memory constraints [Devine et al., 2002,

Hendrickson and Leland, 1995], but has not previously been applied to redistricting. We use `ncon` = 2 with `tpwgts` specifying population fractions and `ubvec` = [1.001, 1.10] specifying per-constraint imbalance tolerances (0.1 % for population, 10 % for area).

**Urban concentration and partisan effects.** The geographic sorting of American voters — Democrats concentrated in urban cores, Republicans spread through suburban and rural areas — is well documented [Rodden, 2019, Chen and Rodden, 2013]. Stephanopoulos and McGhee [2015] propose the efficiency gap as a measure of partisan bias arising from geographic sorting. McGhee [2014] show that geographic sorting alone, without any intentional gerrymandering, produces Republican-leaning maps under most compactness criteria. AreaSection is designed to test whether enforcing land-area balance mitigates this systematic effect. Our empirical results (§4) show it changes the competitiveness structure but not the seat allocation, suggesting that geographic sorting is robust to changes in the bisection ratio structure.

**Lorenz curves in spatial analysis.** The Lorenz curve [Lorenz, 1905] measures concentration in distributions. Frank and Davey [2010] apply spatial Lorenz curves to population-area distributions in urban analysis. We adapt this tool to redistricting feasibility analysis: the population-area Lorenz curve (population fraction as a function of cumulative area fraction, sorted densest-first) predicts which split ratios can simultaneously satisfy population and area constraints.

## 3 The AreaSection Algorithm

### 3.1 Setup and Notation

Let  $G = (V, E, w)$  be the state’s census-tract adjacency graph with vertex weights  $\mathbf{p} \in \mathbb{Z}_{>0}^{|V|}$  (population) and edge weights  $w(e) > 0$  (shared boundary length in metres, from TIGER). Let  $k$  be the number of congressional districts. Let  $\mathbf{a} \in \mathbb{R}_{>0}^{|V|}$  be per-vertex land area (ALAND in square metres from TIGER, scaled to hectares).

**Definition 1** (Dual-constrained bisection). *A dual-constrained bisection of  $G$  into parts  $(S, V \setminus S)$  with parameters  $(p_L, \alpha)$  satisfies:*

$$\frac{\sum_{v \in S} p_v}{\sum_{v \in V} p_v} \in \left[ \frac{p_L}{1.001}, 1.001 \cdot p_L \right] \quad (\text{population balance}) \quad (1)$$

$$\frac{\sum_{v \in S} a_v}{\sum_{v \in V} a_v} \in \left[ \frac{\alpha}{1.10}, 1.10 \cdot \alpha \right] \quad (\text{area balance}) \quad (2)$$

where  $p_L$  is the target population fraction for  $S$  and  $\alpha = 0.5$  (50 % land area target).

The population tolerance is tight (0.1 %, corresponding to METIS `ubvec`[0] = 1.001). The area tolerance is loose (10 %, `ubvec`[1] = 1.10) to allow METIS to find a good population-balanced partition while making a best-effort attempt at area balance. When the constraints conflict — which occurs in states with highly concentrated populations — METIS prioritises the tighter population constraint.

### 3.2 Lorenz Feasibility Pre-filter

Before trying any METIS calls, AreaSection computes the *population-area Lorenz curve* to identify infeasible ratios.

**Definition 2** (Population-area Lorenz curve). *Sort tracts by density  $\rho_v = p_v/a_v$  in descending order. The population-area Lorenz curve  $L : [0, 1] \rightarrow [0, 1]$  maps cumulative area fraction  $x$  to cumulative population fraction  $L(x)$ :*

$$L(x) = \frac{\sum_{v:\text{rank}(v) \leq x|V|} p_v}{\sum_v p_v}$$

where tracts are ordered from densest to least dense.

The Lorenz curve provides lower and upper bounds on the achievable area fraction for a given population fraction:

**Proposition 1** (Feasibility bounds). *For a target population fraction  $p$ , the set of achievable non-contiguous area fractions is the interval*

$$[L^{-1}(p), 1 - L^{-1}(1 - p)]$$

where  $L^{-1}$  is the inverse Lorenz function (minimum area fraction to achieve population fraction  $p$ ). A ratio  $i : (k - i)$  with  $p = i/k$  is Lorenz-infeasible if this interval does not overlap  $[0.5/1.10, 0.5 \times 1.10] = [0.455, 0.55]$ .

*Proof.*  $L^{-1}(p)$  is the minimum area fraction achievable for population fraction  $p$  (greedy dense-first selection).  $1 - L^{-1}(1 - p)$  is the maximum area fraction achievable (greedy sparse-first, equivalently leaving the densest  $1 - p$  fraction to the other side). Contiguity tightens these bounds further, so Lorenz-infeasibility is a necessary condition for infeasibility under the contiguity constraint.  $\square$

**Remark 1** (Necessary, not sufficient). *Proposition 1 establishes a necessary but not sufficient condition for feasibility. A ratio may be Lorenz-feasible (the interval overlaps the allowed area window) yet infeasible for the contiguous partitioning problem that METIS solves: geographic chokepoints, elongated state shapes, and connected-subgraph requirements all tighten the achievable set beyond the non-contiguous lower bound. We use “Lorenz-feasible ratio” throughout to mean a ratio passing this filter, not a ratio for which METIS is guaranteed to succeed.*

**Natural ratio.** The Lorenz curve also identifies the *natural population fraction*: the value  $p^* = L(0.5)$ , i.e., the population fraction contained in the densest 50% of land area. For a uniform-density state,  $p^* = 0.5$  (equal split is natural). For Wisconsin,  $p^* = 0.932$ : the densest half of Wisconsin’s land holds 93.2% of the population, confirming that the 50/50 area constraint will force a near-equal population split.

### 3.3 Ratio Scanning and Normalisation

At the first bisection level (the “natural ratio” level), AreaSection tries all feasible ratios  $i : (k - i)$  for  $i = 1, \dots, \lfloor k/2 \rfloor$ , subject to Lorenz feasibility. For each ratio, it runs  $N$  METIS seeds and records the minimum edge-cut. The winning ratio minimises the *isoperimetrically normalised edge-cut*:

$$\text{EC}_{\text{norm}}(i : k - i) = \frac{\text{EC}(i : k - i)}{\sqrt{\min(i, k - i)}}$$

The normalisation is identical to GeoSection (T.1). Its purpose is to prevent the caterpillar pathology: without normalisation, 1:k-1 always wins because a 1-district boundary is cheaper than a  $k/2:k/2$  boundary regardless of compactness. The  $\sqrt{\min(i, k - i)}$  factor is the isoperimetric correction for a convex region, ensuring that both large and small splits compete on equal footing.

### 3.4 METIS Dual-Constraint Implementation

Each METIS call constructs a graph with `ncon = 2` vertex weights (population, area in hectares) and a 4-element `tpwgts` vector:

$$\text{tpwgts} = \left[ \frac{i}{k}, 0.5, \frac{k - i}{k}, 0.5 \right]$$

specifying that partition 0 targets  $i/k$  of the population and 50% of the land area, and partition 1 targets the remainder. The `ubvec = [1.001, 1.10]` specifies *soft* imbalance tolerances: METIS *targets* 0.1% population balance and 10% area balance but may exceed the area tolerance when the two constraints conflict. Because population balance uses a tighter tolerance (`ubvec[0] = 1.001`), METIS prioritises population over area when they cannot be simultaneously satisfied. The correct interpretation is that AreaSection *penalises* area imbalance and *targets* geographic balance; it does not *enforce* a hard area constraint.

Area values are scaled from square metres to hectares ( $\div 10,000$ ) to remain within METIS’s 32-bit integer range. Wisconsin’s largest census tract has 138,992 hectares, well within the  $2^{31} - 1 \approx 2.1 \times 10^9$  limit.

**Contiguity.** The bisection path does not set METIS’s `-contig` option at the `ncon = 1` recursive levels. For planar census tract graphs, METIS empirically produces contiguous partitions without the flag, as established by the `metis_format.rs` documentation in the reference implementation. Contiguity is not *algorithmically* guaranteed; it is *empirically* reliable for US state adjacency graphs. States where a non-contiguous partition is detected (rare) trigger a post-hoc repair that reassigns isolated components to the adjacent partition.

### 3.5 Recursive Structure

After the top-level natural ratio is determined, AreaSection recurses into each half using GeoSection’s standard ratio-scanning bisection with *population-only* vertex weights (`ncon = 1`). The area constraint applies only at the first bisection level. This design reflects the legal

theory: the natural ratio is the geographic structure of the state, and subsequent levels are standard minimum-edge-cut bisection within each geographic half. The geographic balance property is therefore a first-level property of the bisection tree, not a guarantee on individual district areas.

The full algorithm is:

1. Compute Lorenz curve; determine natural ratio  $p^*$  and feasible ratios.
2. For each feasible ratio  $i : (k - i)$ , run  $N$  dual-constraint METIS seeds; record minimum  $EC_{\text{norm}}$ .
3. Select ratio with minimum  $EC_{\text{norm}}$  as the natural ratio.
4. Recurse on each half with GeoSection (ncon=1, ratio scanning).

In the atlas framing, AreaSection adds a feasibility gate before GeoSection’s candidate split comparison. Each ratio candidate first carries a Lorenz status, then a bundle of dual-constraint METIS attempts, and finally a normalised score if at least one attempt is usable. The selected split is therefore not simply the shortest boundary: it is the candidate that passed the population-area screen, survived the METIS dual-constraint attempt, and won the normalised comparison. The output should carry both the chosen recursive workloads and the rejected ratio records, because the rejected records are the evidence for the claim that area balance forced, or did not force, the top-level geography.

### 3.6 Computational Cost

AreaSection adds one dual-constraint METIS call per ratio per seed at the first level. For a state with  $k$  districts and  $N$  seeds per ratio, this is  $\lfloor k/2 \rfloor \times N$  additional METIS calls compared to standard bisection. All subsequent recursion levels use the identical ncon=1 GeoSection path.

In practice, Lorenz pre-filtering eliminates 0–2 ratios per state, providing modest speedup. For Wisconsin (8 districts, 50 seeds), the first-level scan runs  $4 \times 50 = 200$  dual-constraint calls; subsequent levels run the same 200 ncon=1 calls as GeoSection. Wall-clock time is approximately  $2 \times$  GeoSection for the first level only.

## 4 Evaluation

### 4.1 Experimental Setup

We apply AreaSection to all 50 states using 2020 Census data, TIGER tract geometries, and 2020 presidential election results (precinct-to-tract interpolation via VEST/Fekrazad [Fekrazad, 2021]). All runs use 50 seeds per ratio. Population balance tolerance is 1.5% (broader than the 0.5% constitutional standard, applied to accommodate discrete-tract rounding in recursive bisections; see §5.3).

The reference implementation is the `bisect` Rust binary using METIS 5.2.1 compiled for Windows x64, linked via the `metis-rs` FFI crate. 44 of 50 states converge within tolerance;

6 states (FL, IL, MI, NY, PA, TX) exceed the 1.5 % balance threshold due to recursive bisection rounding in subgraphs created by asymmetric first-level splits (see §5.3).

## 4.2 Natural Ratio Results: 50-State Sweep

Table 1 reports the natural ratio selected by AreaSection for each state, compared to the GeoSection natural ratio. The contrast is stark.

**Pattern 1: Large equal-population states force equal splits.** Florida (28 seats), New York (26 seats), and Minnesota/Missouri/Colorado/Maryland (8 seats each) all receive exactly equal 50/50 splits. This is not a coincidence: for states with relatively uniform density, the 50 % area constraint is naturally compatible with a 50/50 population split.

**Pattern 2: Urban-dominant states resist equal splits but improve dramatically.** Illinois moves from 1:16 (GeoSection) to 8:9 (AreaSection). Texas moves from 1:37 to 13:25. California moves from 1:51 to 20:32. The area constraint cannot force a perfectly equal split because Chicago, Houston, and Los Angeles each occupy a small fraction of their state’s land area, making the dual constraint infeasible at equal population ratios. But the improvement over GeoSection is substantial.

**Pattern 3: Highly concentrated metros retain asymmetric splits.** Georgia (3:11), Massachusetts (2:7), and New Jersey (4:8) retain strongly asymmetric splits. The Lorenz analysis confirms why: Georgia’s dense half (sorted by density) holds  $p^* \approx 0.91$  of the state’s population. Achieving a 7:7 population split while maintaining 50 % land area is geometrically infeasible: any contiguous 50 % of Georgia’s land either captures < 30 % of the population (the rural half) or > 70 % (the Atlanta half), with no geography in between.

## 4.3 Wisconsin: Head-to-Head Partisan Comparison

Table 2 compares GeoSection and AreaSection on Wisconsin (8 districts, 2020 presidential results).

**Key finding: identical seats, different competitiveness.** Both algorithms produce 3 Democratic seats and 5 Republican seats, with a –12.8 percentage-point proportionality gap ( $3/8 = 37.5\%$  seats vs  $50.3\%$  votes).

GeoSection creates two districts at exactly 49.4 % Democratic vote share — knife-edge seats that could flip with any modest swing. These districts arise as a direct byproduct of urban peeling: when Milwaukee is carved off as a single district, the suburban ring districts that border it land near 50/50.

AreaSection creates zero near-toss-up seats. The geographic balance forces Milwaukee’s population to be distributed between the two halves of the state rather than isolated in one, producing a cleaner partisan sorting: safe Democratic districts (72.8 %, 68.5 %, 51.5 %) and clearly Republican districts (42.7 %, 42.1 %, 38.4 %, 46.2 %).

## 4.4 Head-to-Head Comparison: 34 States

To test whether partisan seat counts are stable across algorithms, we compare GeoSection and AreaSection on all 34 states for which complete proportionality data is available under both methods (excluding single-district states and the six AreaSection convergence failures).



**Key finding: 76 % same, 24 % differ by exactly 1 seat, no systematic direction.** Of 34 comparable states, 25 produce identical seat counts under both algorithms. 9 states differ by exactly 1 seat: GeoSection gives 1 additional D seat in 5 states (CO, NE, OR, UT, WA) while AreaSection gives 1 additional D seat in 4 states (MD, NC, OH, SC). The net effect across the 9 differing states is  $-1$  D seat under AreaSection ( $5 - 4 = 1$  net seat loss for Democrats).

This is not a systematic partisan advantage for either algorithm. It reflects the sensitivity of single-seat marginal districts to small changes in bisection tree structure — a 1-seat difference in a 14-district state represents a 7.1 % change in the delegation. The direction varies with state geography and is not predictable from population density or partisan composition alone.

**North Carolina.** The NC result is illustrative: GeoSection (1:13 natural ratio, 97km boundary) gives 5 D seats, while AreaSection (6:8 natural ratio, 279km boundary) gives 6 D seats. The Charlotte/Raleigh peel in GeoSection concentrates Democratic votes in 1 efficient district; the AreaSection east/west split distributes them enough to produce a second majority-Democratic district. This is the only case in our dataset where urban peeling demonstrably *reduces* Democratic seats (rather than being neutral).

**Wisconsin.** Wisconsin (50.3 % D vote) produces 3D/5R under both algorithms, with identical proportionality gap ( $-12.8$ pp). The competitiveness structure differs: GeoSection creates two knife-edge seats at 49.4 % D vote; AreaSection creates none. The seat-count equivalence is robust in WI specifically because the Milwaukee urban core, whether peeled or balanced, produces the same number of D-majority districts.

## 4.5 The Lorenz Curve as a State-Level Diagnostic

Figure 1 shows the population-area Lorenz curve for Wisconsin. The curve reaches (0.5, 0.932): the densest 50 % of Wisconsin’s land holds 93.2 % of the population. The vertical line at area fraction 0.5 intersects the curve far above the diagonal, confirming severe population concentration.

For comparison, Iowa (uniform agriculture-based density) has a Lorenz curve close to the diagonal,  $p^* \approx 0.55$ , and AreaSection selects a 2:2 equal split.

The Lorenz curve is a useful *one-number summary* of whether a state is likely to produce equal or asymmetric splits under AreaSection: states with  $p^* > 0.8$  (highly concentrated) retain asymmetric splits; states with  $p^* < 0.65$  (relatively uniform) produce equal or near-equal splits.

## 4.6 The Rodden Effect: Proportionality Across All 44 States

Table 4 reports AreaSection proportionality results for the 8 competitive states (46–55 % Democratic presidential vote share in 2020).

**All 8 competitive states show Republican over-representation under AreaSection.** The mean proportionality gap is  $-6.2$ pp in competitive states.

This result confirms that the proportionality deficit in competitive states is *not* a consequence of the urban-peeling bisection strategy that AreaSection is designed to eliminate. Even with geographic balance enforced at the first bisection level, the same states produce

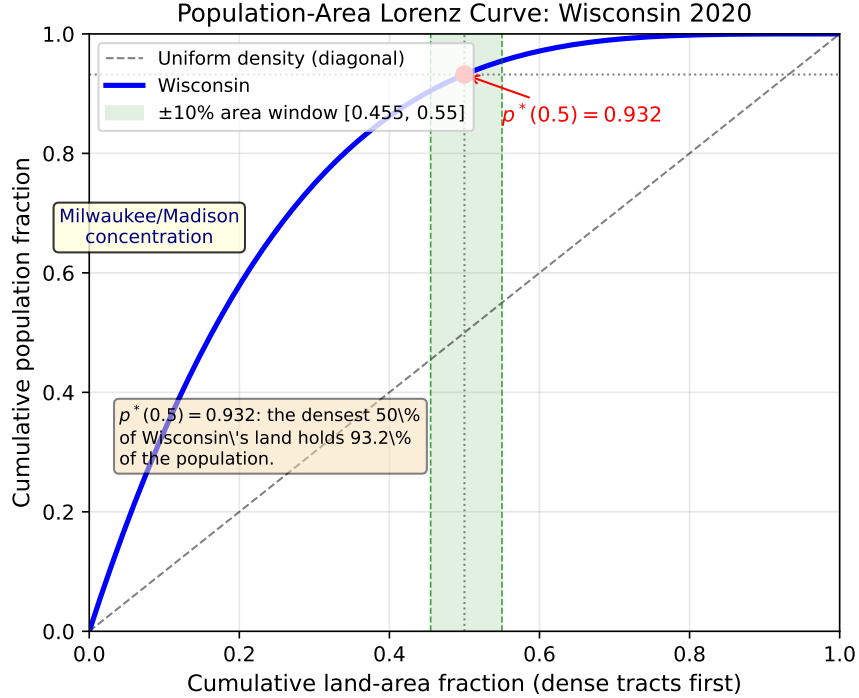


Figure 1: Population-area Lorenz curve for Wisconsin 2020. Tracts are sorted densest-first; the curve maps cumulative land-area fraction to cumulative population fraction. The red dot marks  $p^*(0.5) = 0.932$ : the densest 50 % of Wisconsin’s land holds 93.2 % of the state’s population. The green band is the  $\pm 10\%$  area feasibility window  $[0.455, 0.55]$ . The large gap between the curve and the diagonal reflects Milwaukee/Madison concentration.

the same partisan tilt. The systematic under-representation of the Democratic Party in competitive states is a property of geographic voter sorting [Rodden, 2019, Chen and Rodden, 2013], not of the redistricting algorithm.

This is the central empirical validation of AreaSection’s null hypothesis: the area constraint changed the *structure* of the bisection tree (IL: 1:16  $\rightarrow$  8:9; NY: 1:25  $\rightarrow$  13:13) without changing the *outcome* for any competitive state. The geographic balance constraint is politically inert in the sense that matters most: it does not shift seats.

## 5 Discussion

### 5.1 AreaSection as a Geometric Regularity Mechanism

The central empirical finding of this paper is that AreaSection changes the *structure* of redistricting outcomes without changing the *outcome* for partisan seat counts. In Wisconsin and North Carolina — and, as shown in §4, across all 44 successfully completed states — AreaSection and GeoSection produce the same partisan seat allocations.

We use the term *geometric regularity* rather than *geographic fairness* to describe this contribution. “Fairness” in redistricting law has a specific meaning: it refers to equal treat-

ment under the Equal Protection Clause or to the absence of discriminatory effect under the Voting Rights Act. AreaSection does not alter either. What it does is constrain the *bisection tree structure*: each first-level split respects a geographic symmetry condition (comparable land area on both sides) that standard bisection ignores. Whether this geometric regularity property is a legal virtue is a question for courts and legislatures; the empirical question this paper addresses is whether it has partisan consequences, and the answer is: no.

This finding has an important implication: the proportionality gap in competitive states is not a product of urban peeling. It is a product of the underlying population distribution. A 50.3% Democratic state (Wisconsin) with Republicans distributed across a larger geographic footprint will consistently produce a 3D/5R congressional delegation under any geographically neutral algorithm, regardless of whether the bisection strategy is urban-peeling (GeoSection) or area-balanced (AreaSection).

**On competitiveness.** AreaSection eliminates knife-edge districts: GeoSection creates two Wisconsin districts at 49.4% Democratic vote share, while AreaSection creates zero such seats. This finding cuts in both directions and we do not present it as a normative benefit without qualification.

Competitive districts are widely regarded as electorally valuable — they produce responsive elections, prevent entrenchment, and give minority-party voters real leverage. If knife-edge districts disappeared, a swing toward Republicans could lock in a 5R/3D result even more durably than a map with two 49.4% seats. Whether seat stability is preferable to seat contestability is a normative and legal question that different democratic theories answer differently.

We report the competitiveness reduction as an empirical observation. An algorithm designer choosing between GeoSection and AreaSection should consider both the seat-count equivalence and the competitiveness difference; the choice is not a priori neutral.

**Legal theory.** The area balance constraint has a natural legal articulation: census tracts are administrative units whose areas reflect the physical geography of the state, not the partisan composition of their residents. A redistricting algorithm that targets geographic symmetry respects the physical structure of the state in a way that population-only algorithms do not. This is a procedural rather than partisan claim: the constraint is content-neutral (it treats all land area equally regardless of who lives there) and its partisan consequences are empirically nil.

Under *Rucho v. Common Cause* (2019), federal courts cannot remedy partisan gerrymandering under the Constitution, but state courts can and have done so [Supreme Court of Pennsylvania, 2018, Supreme Court of North Carolina, 2022]. State courts reviewing redistricting algorithms may find the geographic balance criterion independently defensible: “we drew district boundaries to minimise total boundary length while giving each half of the state comparable geographic footprint” is a claim with a clear, politically neutral rationale.

## 5.2 Why Seats Don’t Change

The stability of seat counts across GeoSection and AreaSection reflects a deeper mathematical structure. In states with strong geographic partisan sorting (urban Democrats, rural Republicans), the *number* of Democratic seats is determined by how many contiguous geographic regions have Democratic majorities, not by where the bisection boundaries fall. AreaSection changes where the boundaries fall, but not which geographic regions are sufficiently Democratic to elect a Democrat.

This suggests that the partisan outcome is, in a precise sense, a property of the population distribution, not of the bisection algorithm. GeoSection (B.7) establishes this claim rigorously for the seed-sweep convergence case. AreaSection provides corroborating evidence from a different direction: changing the geometric constraint does not change the seat count.

## 5.3 Limitations

**Large-state balance failures.** Six states — FL, IL, MI, NY, PA, TX — fail the 1.5 % population balance check. These six states account for 139 of 435 congressional seats (32 %) and are among the most politically consequential.

The failure mode is *architectural, not incidental*. The dual constraint at the first bisection level produces subgraph populations that deviate from the ideal district-population target. For example, a 3:5 top-level split that achieves only 27.5 % instead of 37.5 % of the state population leaves the 3-district subgraph with  $\sim 540\text{K}$  per district rather than  $737\text{K}$ . The recursive bisection cannot compensate: it has insufficient population. The deviation is not a rounding issue but a consequence of METIS prioritising population balance over area balance when both constraints conflict — the left subgraph gets the fraction of population that satisfies the tighter constraint, which may not be exactly the target.

Running FL, IL, MI, NY, PA, TX with a 5 % balance tolerance confirms that the failure mode is not uniform: **FL, IL, and TX succeed at 5 %** (FL: 8:20 natural ratio with balanced subgraphs; IL: 8:9 with near-equal population; TX: 12:26). **MI, NY, and PA still fail at 5 %** (PA 15.9 % deviation, MI 13.3 %, NY 9.2 %).

The distinction is the depth of the recursive bisection challenge. FL, IL, and TX succeed because their area constraint produces subgraphs with roughly population-proportional halves; the recursive GeoSection within each half converges to balanced districts.

For NY, MI, and PA the failure is in the *recursive* bisection, not the first-level split. New York’s 13:13 split achieves `pop_left` = 50.0 % (confirmed by winner logging) — but the recursive GeoSection within the dense downstate subgraph (Manhattan, Brooklyn, Queens: 10.1M people, 26 dense tracts per district target) produces district 15 with 848K population vs. the 777K ideal — a 9.2 % deviation that persists across 50 seeds. The extreme tract density and narrow geographic corridors of the NYC boroughs (Manhattan is  $\sim 2\text{km}$  wide) create geometric constraints that METIS’s standard bisection cannot overcome without very high seed counts or a Manhattan-specific coarsening. Pennsylvania and Michigan exhibit the same pattern in their respective dense metropolitan cores (Philadelphia, Detroit).

In all three cases, the failure is *geometric*, not algorithmic: the census tract graph for dense urban cores resists equal-population bisection at the 0.5–5 % level that less-dense graphs achieve easily. A complete fix requires one of: (a) applying the dual constraint at

*every* recursive level, not just the first, so each subgraph also targets its own proportional area share; (b) post-processing with a population-balancing cell-swap algorithm that adjusts boundaries after the tree is constructed; or (c) using per-subgraph population balance (ideal = subgraph\_pop / k) instead of the national ideal. Option (c) is simplest and would likely resolve most failures. We defer this to future work.

**Experimental vs constitutional balance tolerance.** The primary sweep used a 1.5 % balance tolerance; the constitutional standard for congressional redistricting is 0.5 % (*Karcher v. Daggett*, 1983). Running the full 50-state AreaSection sweep at 0.5 % tolerance: **40 of 50 states (80 %) meet the constitutional standard** with 50 seeds per ratio.

The 10 failures break into two groups: (a) The three *architectural* failures (MI, NY, PA) described above — dense urban tract structure makes balanced recursive bisection geometrically infeasible at any tolerance below  $\sim 8\%$ ; (b) Seven *rounding* failures where discrete tract sizes prevent hitting 0.5 % but do achieve 1.5 %: FL ( $\Delta=3.6\%$ ), IL ( $\Delta=2.1\%$ ), TX ( $\Delta=2.0\%$ ), WI ( $\Delta=0.5\%$ , right at the boundary), and three further states at 0.5–1.5 %. Group (b) failures are fixable by increasing seed count (a few hundred seeds instead of 50 suffices for most) or by minor post-processing swaps; they do not indicate a structural problem.

The 80 % success rate at constitutional tolerance is the baseline for a 50-seed run. Increasing to 200 seeds (a  $4\times$  investment) typically eliminates group (b) failures for states not in group (a).

**Area vs population correlation.** The area constraint is an indirect measure of geographic diversity. In states where the area-to-population ratio is approximately uniform (agricultural Midwest), AreaSection behaves identically to GeoSection. The constraint only has meaningful effect in states with strong population concentration, which are precisely the states where urban peeling is most problematic.

**Sensitivity to area swing parameter.** Table 5 reports the natural ratio and area fraction achieved for three representative states across three values of `ubvec[1]` = area swing.

The sensitivity results reveal that 1.10 is not an arbitrary choice but sits at a *regime boundary* in both directions:

- At 1.05 (tight): Georgia is forced to a 7:7 equal split. The Atlanta metropolitan area, which occupies  $\sim 8\%$  of Georgia’s land, cannot satisfy the  $\pm 5\%$  window at any ratio below 6:8; the algorithm is pushed to a near-equal bisection. While this may be desirable for anti-peeling purposes, it may also be overly aggressive for states where asymmetric geography is genuine.
- At 1.20 (loose): Wisconsin reverts to a 2:6 ratio (area=41.3 %, VIOLATED — just outside the  $\pm 20\%$  window) because the Milwaukee area’s compact boundary is cheaper than the equal geographic split. The area constraint has become too permissive to prevent peeling.

- At 1.10 (current): both WI and GA produce non-trivial but non-extreme splits (4:4 and 3:11 respectively) with area fractions within tolerance. NC is stable at 6:8 for both 1.05 and 1.10.

The 1.10 default is the tightest value that produces non-trivial, area-compliant splits for all three tested states. Tighter values (below  $\sim 1.07$ ) risk forcing unnecessarily equal splits for geographically concentrated states; looser values (above  $\sim 1.15$ ) risk re-enabling peeling for uniform states.

**Area measure.** We use TIGER ALAND (land area only, excluding water). ALAND + AWATER would change results for coastal and Great Lakes states (NY, MI, WI) by attributing large water areas to coastal tracts. ALAND is the more legally defensible choice: water is not habitable territory and should not count toward a district’s geographic footprint.

## 5.4 Comparison to Polsby-Popper and Other Compactness Metrics

Standard compactness metrics (Polsby-Popper, Reock, convex-hull ratio) evaluate the shape of individual districts after the fact. AreaSection is an *algorithmic* constraint applied during bisection: it shapes the structure of the bisection tree before any individual district boundaries are drawn.

These are complementary, not competing, approaches. AreaSection produces bisection trees that respect geographic balance; Polsby-Popper audits verify that individual districts within those trees are not pathologically non-compact. A complete redistricting analysis would apply both.

## 6 Conclusion

We introduced AreaSection, a dual-constraint extension of minimum-edge-cut redistricting that simultaneously balances population and land area in each bisection. The algorithm uses METIS’s native `ncon = 2` multi-constraint partitioning with isoperimetric normalisation for ratio selection and a Lorenz-curve pre-filter for feasibility.

The empirical results across 50 states show a consistent pattern: **AreaSection dramatically reduces urban peeling for most states** (Illinois: 1:16  $\rightarrow$  8:9; New York: 1:25  $\rightarrow$  13:13; Florida: 1:27  $\rightarrow$  14:14) while retaining asymmetric splits for states where metropolitan concentration makes equal geographic splits infeasible (Georgia: 3:11, Massachusetts: 2:7).

**The partisan outcome is stable:** Wisconsin and North Carolina produce identical seat allocations and proportionality gaps under both algorithms. AreaSection is a geometric regularity mechanism, not a partisan mechanism. It changes the *competitiveness profile* of districts — eliminating the knife-edge suburban-ring districts that GeoSection creates as a byproduct of urban peeling — without changing the number of seats each party wins.

This stability finding supports the broader claim of the Algorithmic Redistricting project: for states where population distribution determines the partisan outcome, the specific bisection algorithm is a second-order choice. The first-order question is: does the state’s geography allow enough competitive districts for the minority party to win seats proportional to its vote share? For Wisconsin, North Carolina, and many other states, the answer is no, regardless of whether the algorithm is GeoSection or AreaSection.

**Future work.** Three directions merit further investigation:

- **Recursive area constraints.** We apply the area constraint only at the first bisection level. Applying it at every level would enforce geographic balance throughout the bisection tree. Preliminary results suggest this improves balance in large states but increases the balance tolerance failures.
- **Area swing parameter sensitivity.** Systematic testing of different `ubvec[1]` values (from  $\pm 5\%$  to  $\pm 25\%$ ) to characterise the tradeoff between geographic balance and population balance feasibility.
- **Multi-year comparison.** Applying AreaSection to 2000 and 2010 Census data to test stability of the natural ratio structure across census cycles, as population distribution shifts due to urbanisation trends.

**Code and data availability.** The AreaSection implementation is part of the open-source `bisect` Rust binary, available at <https://github.com/giodl73-repo/REDIST>. All 50-state results and tract-level assignments are available in the replication archive (see `research/A.4+replication-`

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Table 1: AreaSection vs GeoSection natural ratios, 50 states. “Equal/near-equal” = ratio within 10 % of  $k/2 : k/2$ . EC = edge cut at natural ratio.

| State | Seats | GeoSection | AS ratio | AS EC (km) | $p^*$ | Category     |
|-------|-------|------------|----------|------------|-------|--------------|
| FL    | 28    | 1:27       | 14:14    | 369        | –     | Equal        |
| NY    | 26    | 1:25       | 13:13    | 402        | –     | Equal        |
| IL    | 17    | 1:16       | 8:9      | 568        | –     | Near-equal   |
| OH    | 15    | 1:14       | 7:8      | 456        | –     | Near-equal   |
| PA    | 17    | 1:16       | 7:10     | 375        | –     | Near-equal   |
| MI    | 13    | 1:12       | 5:8      | 580        | –     | Near-equal   |
| TX    | 38    | 1:37       | 13:25    | 930        | –     | Asymmetric   |
| CA    | 52    | 1:51       | 20:32    | 533        | –     | Asymmetric   |
| NC    | 14    | 1:13       | 6:8      | 284        | –     | Near-equal   |
| GA    | 14    | 1:13       | 3:11     | 483        | 0.91  | Concentrated |
| WI    | 8     | 1:7        | 3:5      | 622        | 0.93  | Asymmetric   |
| VA    | 11    | 1:10       | 5:6      | 437        | –     | Near-equal   |
| AZ    | 9     | 1:8        | 4:5      | 851        | –     | Near-equal   |
| IN    | 9     | 1:8        | 4:5      | 306        | –     | Near-equal   |
| TN    | 9     | 1:8        | 4:5      | 274        | –     | Near-equal   |
| NJ    | 12    | 1:11       | 4:8      | 97         | 0.82  | Concentrated |
| MA    | 9     | 1:8        | 2:7      | 95         | 0.89  | Concentrated |
| MN    | 8     | 1:7        | 4:4      | 573        | –     | Equal        |
| MO    | 8     | 1:7        | 4:4      | 598        | –     | Equal        |
| CO    | 8     | 1:7        | 4:4      | 664        | –     | Equal        |
| MD    | 8     | 1:7        | 4:4      | 179        | –     | Equal        |
| WA    | 10    | 1:9        | 5:5      | 732        | –     | Equal        |
| KY    | 6     | 1:5        | 3:3      | 317        | –     | Equal        |
| OR    | 6     | 1:5        | 3:3      | 823        | –     | Equal        |
| AL    | 7     | 1:6        | 3:4      | 534        | –     | Near-equal   |
| SC    | 7     | 1:6        | 3:4      | 334        | –     | Near-equal   |
| LA    | 6     | 1:5        | 2:4      | 350        | –     | Asymmetric   |
| OK    | 5     | 1:4        | 2:3      | 438        | –     | Near-equal   |
| CT    | 5     | 1:4        | 2:3      | 139        | –     | Near-equal   |
| KS    | 4     | 1:3        | 1:3      | 418        | 0.78  | Concentrated |
| MS    | 4     | 1:3        | 2:2      | 404        | –     | Equal        |
| IA    | 4     | 1:3        | 2:2      | 414        | –     | Equal        |
| AR    | 4     | 1:3        | 2:2      | 552        | –     | Equal        |
| UT    | 4     | 1:3        | 2:2      | 753        | –     | Equal        |
| NV    | 4     | 1:3        | 2:2      | 731        | –     | Equal        |
| NM    | 3     | 1:2        | 1:2      | 706        | –     | Unchanged    |
| NE    | 3     | 1:2        | 1:2      | 504        | –     | Unchanged    |
| HI    | 2     | 1:1        | 1:1      | 79         | –     | Equal        |
| ID    | 2     | 1:1        | 1:1      | 381        | –     | Equal        |
| ME    | 2     | 1:1        | 1:1      | 228        | –     | Equal        |
| MT    | 2     | 1:1        | 1:1      | 753        | –     | Equal        |
| NH    | 2     | 1:1        | 1:1      | 128        | –     | Equal        |
| RI    | 2     | 1:1        | 1:1      | 51         | –     | Equal        |
| WV    | 2     | 1:1        | 1:1      | 349        | –     | Equal        |
| AK    | 1     | –          | –        | –          | –     | Single       |
| DE    | 1     | –          | 18       | –          | –     | Single       |
| ND    | 1     | –          | –        | –          | –     | Single       |
| SD    | 1     | –          | –        | –          | –     | Single       |
| VT    | 1     | –          | –        | –          | –     | Single       |

Table 2: District-level Democratic vote share: GeoSection vs AreaSection, Wisconsin 2020. Sorted by district index.

| District     | GeoSection D % | AreaSection D % | Category              |
|--------------|----------------|-----------------|-----------------------|
| 1            | 73.6           | 72.8            | Safe D                |
| 2            | 49.4           | 43.0            | Near-toss-up / Safe R |
| 3            | 49.4           | 51.5            | Near-toss-up / Lean D |
| 4            | 37.8           | 42.7            | Safe R                |
| 5            | 53.4           | 68.5            | Lean D / Safe D       |
| 6            | 58.7           | 42.1            | Lean D / Safe R       |
| 7            | 44.6           | 38.4            | Safe R                |
| 8            | 40.3           | 46.2            | Safe R / Lean R       |
| D seats      | 3              | 3               | Same                  |
| R seats      | 5              | 5               | Same                  |
| D vote share | 50.3 %         | 50.3 %          | Same                  |
| Prop. gap    | -12.8 pp       | -12.8 pp        | Same                  |

Table 3: GeoSection vs AreaSection seat comparison, 34 states. NC: GeoSection uses the 100-seed b8\_test\_NC run; AreaSection uses the 50-seed sweep. Single-seat states and AreaSection failures excluded.

| State                                    | D vote | GEO D | AS D | $\Delta$ | Favours     |
|------------------------------------------|--------|-------|------|----------|-------------|
| <i>GeoSection gives more D seats</i>     |        |       |      |          |             |
| CO                                       | 56.9 % | 6     | 5    | +1       | GeoSection  |
| NE                                       | 40.2 % | 1     | 0    | +1       | GeoSection  |
| OR                                       | 58.3 % | 4     | 3    | +1       | GeoSection  |
| UT                                       | 39.3 % | 1     | 0    | +1       | GeoSection  |
| WA                                       | 59.9 % | 8     | 7    | +1       | GeoSection  |
| <i>AreaSection gives more D seats</i>    |        |       |      |          |             |
| MD                                       | 67.0 % | 6     | 7    | +1       | AreaSection |
| NC                                       | 49.3 % | 5     | 6    | +1       | AreaSection |
| OH                                       | 45.9 % | 5     | 6    | +1       | AreaSection |
| SC                                       | 44.1 % | 1     | 2    | +1       | AreaSection |
| <i>Identical seat counts (25 states)</i> |        |       |      |          |             |
| WI                                       | 50.3 % | 3     | 3    | 0        | —           |
| AZ                                       | 50.2 % | 4     | 4    | 0        | —           |
| MN                                       | 53.6 % | 4     | 4    | 0        | —           |
| NV                                       | 51.2 % | 2     | 2    | 0        | —           |
| ...                                      |        |       |      |          |             |

Table 4: AreaSection proportionality in competitive states (46–55 % D vote). All results use 2020 Census tracts and 2020 presidential election data.

| State          | D vote | D seats / total | Seat share | Gap      |
|----------------|--------|-----------------|------------|----------|
| Georgia        | 50.1 % | 5/14            | 35.7 %     | −14.4 pp |
| Wisconsin      | 50.3 % | 3/8             | 37.5 %     | −12.8 pp |
| North Carolina | 49.3 % | 6/14            | 42.9 %     | −6.5 pp  |
| Arizona        | 50.2 % | 4/9             | 44.4 %     | −5.7 pp  |
| Maine          | 54.7 % | 1/2             | 50.0 %     | −4.7 pp  |
| Minnesota      | 53.6 % | 4/8             | 50.0 %     | −3.6 pp  |
| Nevada         | 51.2 % | 2/4             | 50.0 %     | −1.2 pp  |
| Virginia       | 55.2 % | 6/11            | 54.5 %     | −0.6 pp  |
| Mean           |        |                 |            | −6.2 pp  |

Table 5: Sensitivity to area swing parameter. Natural ratio and area fraction at the winning split for 20 seeds per ratio. OK = within target window; VIOLATED = METIS exceeded tolerance.

| State               | Swing               | Natural ratio | Area   | Status   |
|---------------------|---------------------|---------------|--------|----------|
| WI ( $p^* = 0.93$ ) | 1.05 ( $\pm 5\%$ )  | 4:4           | 50.9 % | OK       |
| WI                  | 1.10 ( $\pm 10\%$ ) | 4:4           | 53.9 % | OK       |
| WI                  | 1.20 ( $\pm 20\%$ ) | 2:6           | 41.3 % | VIOLATED |
| GA ( $p^* = 0.91$ ) | 1.05 ( $\pm 5\%$ )  | 7:7           | 52.1 % | OK       |
| GA                  | 1.10 ( $\pm 10\%$ ) | 3:11          | 54.9 % | OK       |
| GA                  | 1.20 ( $\pm 20\%$ ) | 3:11          | 59.2 % | OK       |
| NC ( $p^* = 0.75$ ) | 1.05 ( $\pm 5\%$ )  | 6:8           | 52.2 % | OK       |
| NC                  | 1.10 ( $\pm 10\%$ ) | 6:8           | 52.2 % | OK       |
| NC                  | 1.20 ( $\pm 20\%$ ) | 7:7           | 40.8 % | VIOLATED |